

Stack Control in Motorola 680X0

Definition of a Stack

A **stack** is a sequence of storage locations that are accessible only from one end, known as the **top of the stack**. It follows a **Last-In-First-Out (LIFO)** structure.

Basic Stack Operations

- **Push:** Inserts an operand onto the top of the stack.
- **Pop:** Removes the operand from the top of the stack.

Both operations modify the **Stack Pointer (SP)**, which points to the current top of the stack.

Stack Behavior

- **Push:** Decrements the stack pointer before storing the operand.
- **Pop:** Reads the operand before incrementing the stack pointer.
- The change in SP depends on operand size (e.g., 4 bytes for a long word).

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The Motorola 680X0 series lacks dedicated stack hardware but supports stack-like behavior through its **flexible addressing modes**.

Any address register (e.g., **A2**) can act as a stack pointer. The stack grows toward **lower memory addresses**.

Push Operation Example

To push the contents of data register **D6** onto the stack:

MOVE.L D6, -(A2)

Equivalent low-level steps:

$$A2 := A2 - 4, \quad M[A2] := D6$$

Pop Operation Example

To pop the top of the stack into D6:

MOVE.L (A2)+, D6

Equivalent low-level steps:

$$D6 := M[A2], \quad A2 := A2 + 4$$

Stack Operation Illustration

Before pushing:

- $A2 = 0x1000$

After `MOVE.L D6, -(A2)`:

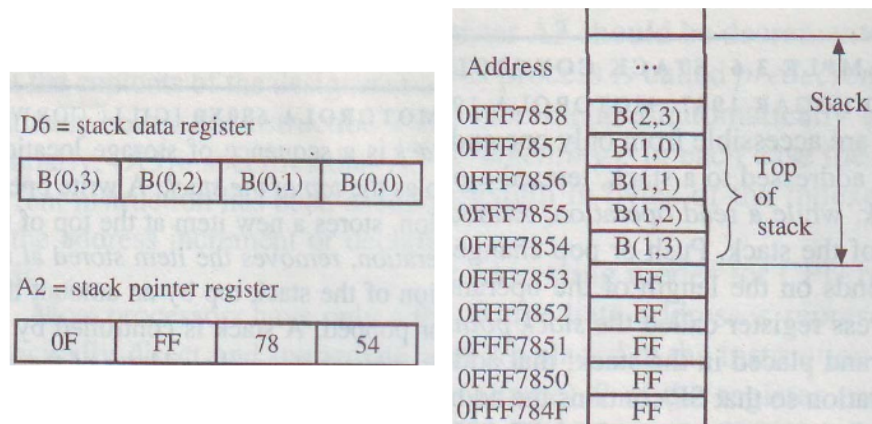
- $A2 = 0x0FFC$
- Memory at $0x0FFC$ stores value of D6

After `MOVE.L (A2)+, D6`:

- D6 is loaded from $0x0FFC$
- $A2 = 0x1000$

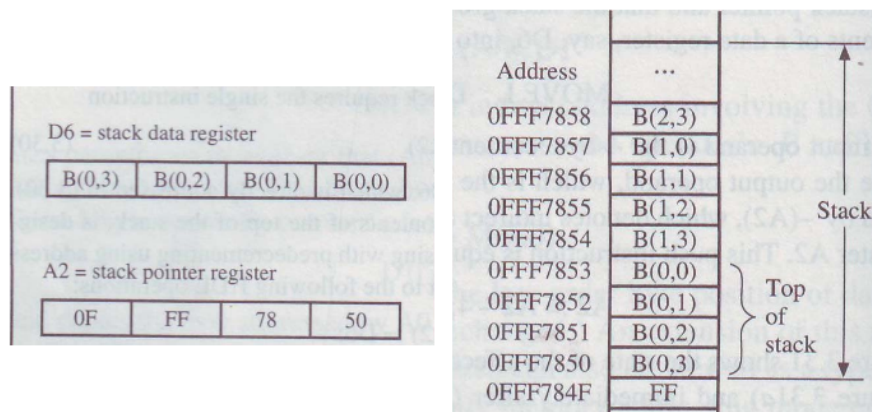
Diagram (Optional)

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Prior the execution of `MOVE.L D6, -(A2)`

Stack Control in Motorola 680X0



After the execution of `MOVE.L D6, -(A2)`

(The diagrams demonstrate the push and pop operations visually with memory layout and register behavior.)